System of Linear Equation

Non-Homogeneous

$AX=B$

$$\underset{A}{\begin{bmatrix} 2 & 3 & 1 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}} \underset{X}{\begin{bmatrix} x \\ y \\ z \end{bmatrix}} = \underset{B}{\begin{bmatrix} 7 \\ 5 \\ 10 \end{bmatrix}}$$

Homogeneous

$AX = 0$

$$\underset{A}{\begin{bmatrix} 5 & 6 & 1 \\ 4 & 3 & 2 \\ 3 & 1 & 5 \end{bmatrix}} \underset{X}{\begin{bmatrix} x \\ y \\ z \end{bmatrix}} = \underset{B}{\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}}$$

$\det(A) \neq 0$ [Unique Solution & Consistent]

$\det(A) = 0$ [Non-trivial/No Solution & Inconsistent]

Q. What is the name of the following linear system of equations? Solve it & identify the type of Solution

$$x+y-z=0, \quad 2x+4y-z=0 \quad \text{and} \quad 3x+2y+2z=0$$

$\Rightarrow$ Given that,

$x+y-z=0$ —— L_1

$2x+4y-z=0$ —— L_2

$3x+2y+2z=0$ —— L_3

$$\underset{A}{\begin{bmatrix} 1 & 1 & -1 \\ 2 & 4 & -1 \\ 3 & 2 & 2 \end{bmatrix}} \underset{X}{\begin{bmatrix} x \\ y \\ z \end{bmatrix}} = \underset{0}{\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}}$$

Hence, the system of equation is Homogeneous [$\because AX=0$] (Ans)

We know that,

$$L_i = -a_{i1}L_1 + a_{11}L_i \quad [i>1]$$

Now putting $i=2$ we get,

$$L_2 = -a_{21} L_1 + a_{11} L_2$$
$$= -2(x+y-z) + 1(2x+4y-z)$$
$$= -2x-2y+2z+2x+4y-z$$
$$= 2y+z$$

Again putting $i=3$ we get,

$$L_3 = -a_{31} L_1 + a_{11} L_3$$
$$= -3(x+y-z) + 1(3x+2y+2z)$$
$$= -3x-3y+3z+3x+2y+2z$$
$$= -y+5z$$

Thus we obtains the following equations

$$x+y-z=0$$
$$2y+z=0 \quad \text{——} \; L_1$$
$$-y+5z=0 \quad \text{——} \; L_2$$

Now putting $i=2$ we get,

$$L_2 = -a_{21} L_1 + a_{11} L_2$$
$$= -(-1)(2y+z) + 2(-y+5z)$$
$$= 2y+z-2y+10z = 11z$$

Thus, we obtain the following equations

$$x + y - z = 0$$

$$2y + z = 0$$

$$11z = 0$$

Now,

$$11z = 0 \quad \therefore z = 0$$

Again,

$$2y + 0 = 0 \quad \therefore y = 0$$

Also,

$$x + 0 - 0 = 0 \quad \therefore x = 0$$

Hence $(x, y, z) = (0, 0, 0)$ // (Ans)

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 4 & -1 \\ 3 & 2 & 2 \end{bmatrix}$$

$$\therefore \det(A) = 1(8+2) - 1(4+3) - 1(4-12)$$

$$= 10 - 7 + 8 = 11 \neq 0$$

$\det(A) \neq 0$.

It has an Unique Solution & Consistent. // (Ans)

Solve the system of linear equations by $L_i \to -a_{i1}L_1 + a_{11}L_i$

$$2x - y + 3z = 9$$
$$x - 3y \cancel{+} - 2z = 0$$
$$3x + 2y - z = -1$$

⇨ Given that,

$$2x - y + 3z = 9 \quad \text{—— } L_1$$
$$x - 3y - 2z = 0 \quad \text{—— } L_2$$
$$3x + 2y - z = -1 \quad \text{—— } L_3$$

We know that,

$$L_i = -a_{i1}L_1 + a_{11}L_i \qquad [i > 1]$$

Now putting $i = 2$ we get,

$$L_2 = -a_{21}L_1 + a_{11}L_2$$
$$= -1(2x - y + 3z - 9) + 2(x - 3y - 2z)$$
$$= -2x + y - 3z + 9 + 2x - 6y - 6z$$
$$= -5y - 9z + 9$$

Again putting $i = 3$ we get,

$$L_3 = -a_{31}L_1 + a_{11}L_3$$
$$= -3(2x - y + 3z - 9) + 2(3x + 2y - z + 1)$$
$$= -6x + 3y - 9z + 9 + 6x + 4y - 2z + 2$$

$= 7y - 11z + 11$

Thus we obtain the following equations.

$2x - y + 3z = 9$

$-5y - 9z + 9 = 0$ —— L_1

$7y - 11z + 11 = 0$ —— L_2

Now putting $i = 2$ we get,

$L_2 = -a_{21} L_1 + a_{11} L_2$

$= -7(-5y - 9z + 9) + (-5)(7y - 11z + 11)$

$= 35y + 63z - 63 - 35y + 55z - 55$

$= 118z - 118$

Thus we obtain the following equations,

$2x - y + 3z = 9$

$-5y - 9z + 9 = 0$

$118z - 118 = 0$

Now,

$118z = 118 \quad \therefore z = 1$

Again, $-5y - 9 \times 1 + 9 = 0 \quad \therefore y = 0$

Also, $2x - 0 + 3 \times 1 = 9 \quad \therefore x = 3$

Hence $(x, y, z) = (3, 0, 1)$ // (Ans)

Using Matrix method, Linear equation

Q. Using matrix method, solve the following equations,

$x+y+z=6,\ x-y+z=2$ and $2x+y-z=7$

$\Rightarrow$ Given that,

$$x+y+z=6$$
$$x-y+z=2$$
$$2x+y-z=1$$

let,

$$A=\begin{bmatrix}1&1&1\\1&-1&1\\2&1&-1\end{bmatrix},\quad X=\begin{bmatrix}x\\y\\z\end{bmatrix}\quad \text{and}\quad K=\begin{bmatrix}6\\2\\1\end{bmatrix}$$

Such that,

$$AX=K$$

$$\text{or, } A^{-1}.A.X=A^{-1}K \qquad [\because A.A^{-1}=I]$$

$$\therefore X=A^{-1}.K \quad \text{---(1)}$$

Now,

$$|A|=\begin{vmatrix}1&1&1\\1&-1&1\\2&1&-1\end{vmatrix}=1(1-1)-1(-1-2)+1(1+2)$$
$$=6$$

Again the cofactor of the matrix of A is given by,

$$C_{11}=\begin{vmatrix}-1&1\\1&-1\end{vmatrix} \qquad C_{12}=-\begin{vmatrix}1&1\\2&-1\end{vmatrix} \qquad C_{13}=\begin{vmatrix}1&-1\\2&1\end{vmatrix}$$

$$=1-1=0 \qquad =-(-1-2)=3 \qquad =1+2=3$$

$C_{21} = -\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}$ $\qquad C_{22} = \begin{vmatrix} 1 & 1 \\ 2 & -1 \end{vmatrix}$ $\qquad C_{23} = -\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix}$

$= -(-1-1) = 2 \qquad = -1-2 = -3 \qquad = -(1-2) = 1$

$C_{31} = \begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix}$ $\qquad C_{32} = -\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = -(1-1) = 0$ $\qquad C_{33} = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}$

$= 1+1 = 2 \qquad\qquad = -1-1 = 2$

So, the co-factor matrix of A is given by.

$$A^c = \begin{bmatrix} 0 & 3 & 3 \\ 2 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$$

$$\therefore \text{Adj of } A = (A^c)^t = \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 1 \\ 3 & 1 & -2 \end{bmatrix}$$

Hence, $A^{-1} = \dfrac{\text{Adj of } A}{|A|} = \dfrac{1}{6} \times \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 1 \\ 3 & 1 & -2 \end{bmatrix}$

$$= \begin{bmatrix} 0 & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{2} & \frac{-1}{2} & 0 \\ \frac{1}{2} & \frac{1}{6} & -\frac{1}{3} \end{bmatrix}$$

Now from eqn (1) $\Rightarrow$

$$X = A^{-1} K$$

$$= \begin{bmatrix} 0 & 1/3 & 1/3 \\ 1/2 & -1/2 & 0 \\ 1/2 & 1/6 & -1/3 \end{bmatrix} \times \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 + \frac{1}{3}\times 2 + \frac{1}{3}\times 1 \\ \frac{1}{2}\times 6 + \left(-\frac{1}{2}\right)\times 2 + 0 \\ \frac{1}{2}\times 6 + \frac{1}{6}\times 2 + \left(-\frac{1}{3}\right)\times 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{2+1}{3} \\ 3-1+0 \\ 3+\frac{1}{3}-\frac{1}{3} \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$\therefore x = 1, \; y = 2, \; z = 3$ // Ans

Cayley – Hamilton Theorem

4. State Cayley – Hamilton. Verify the Cayley – Hamilton theorem hence find A^{-1} for $A = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$

⇨ Statement: Every Square matrix satisfies its characteristic equation, OR.

If $|A - \lambda I| = (-1)^n \left[\lambda^n + a_1\lambda^{n-1} + a_2\lambda^{n-2} + \cdots a_n\right]$ be the characteristic polynomial of $n\times n$ matrix. $A = [a_{ij}]$, then the matrix equation

$X_n + a_2 X^{n-1} + \cdots + a_n I = 0$ is satisfied by $X = A$

$$\boxed{A^n + a_1 . A^{n-1} + \cdots + a_n I = 0}$$

Given that,

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$$

Now, $A - \lambda I = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$= \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} 1-\lambda & 2 \\ 1 & 1-\lambda \end{bmatrix}$$

$\therefore$ The characteristic equation of A,

$$|A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 2 \\ 1 & 1-\lambda \end{vmatrix} = 0$$

or, $(1-\lambda)^2 - 2 = 0$

or, $1 - 2\lambda + \lambda^2 - 2 = 0$

or, $\lambda^2 - 2\lambda - 1 = 0$

Now, by the Cayley Hamilton theorem we ~~got~~ have,

$$A^2 - 2A - 1 = 0 \quad \text{——(i)}$$

Now, $A^2 = A \times A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1+2 & 2+2 \\ 1+1 & 2+1 \end{bmatrix}$

$$= \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$$

Now from eqn ① $\Rightarrow$

$$L.S = A^2 - 2A - I$$

$$= \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix} - 2\times\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 4 \\ 2 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3-2-1 & 4-4-0 \\ 2-2-0 & 3-2-1 \end{bmatrix} = 0 = R.S$$

Hence, Cayley Hamilton's theorem is verified.

We have,

$$A^2 - 2A - I = 0$$

Now multiplying A^{-1} we get

$$A^2 . A^{-1} - 2A . A^{-1} - I A^{-1} = 0$$

$$\text{or, } A^{-1} = A - 2I$$

$$= \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} - 2\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1-2 & 2-0 \\ 1-0 & 1-2 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix} \text{ // (Ans)}$$

Eigen Vector & Eigen Values

Defination:

An Eigen vector, is a non-zero which satisfies the equation, $A\vec{V} = \lambda\vec{V}$ or, $A\vec{X} = \lambda\vec{X}$

where,

A is the given square matrix

λ is the scalar value which is called the eigen value of the given matrix,

$\vec{V}$ or $\vec{X}$ is called the eigen vector of the given matrix.

Q. Check whether the vector $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$ are eigen vector for $A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}$. Showing your analysis procedure graphically.

⇨ We know the relation between Eigen vector and Eigen values of a matrix, $A\vec{X} = \lambda\vec{X}$

where,

A is given matrix

λ is Eigen values of the given matrix

$\vec{X}$ is the corresponding Eigen vectors of matrix

P. T. O

a) <u>Mathematical Analysis:</u> for $\vec{X} = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$

Given that,

$$A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}$$

Let, $\vec{X} = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$

$$A\vec{X} = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 6 \end{pmatrix} = \begin{pmatrix} 4+12 \\ -16+42 \end{pmatrix} = \begin{pmatrix} 16 \\ 26 \end{pmatrix}$$

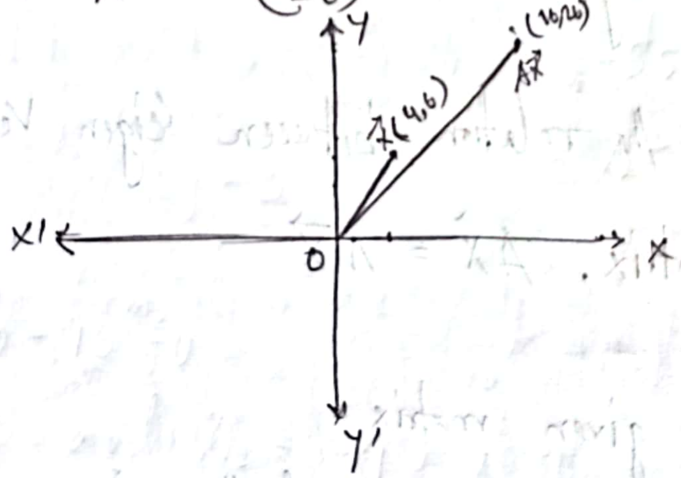
$$= 2\begin{pmatrix} 8 \\ 13 \end{pmatrix}$$

As $A\vec{X} = \lambda\vec{X}$ is not satisfied. So, the given vector, $\vec{X} = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$ is not eigen vector.

//Ans

a) <u>Graphical Analysis:</u>

Here, $A\vec{X} = \begin{pmatrix} 16 \\ 26 \end{pmatrix}$ and $\vec{X} = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$



As the product vector, $A\vec{X}$ is not in the same direction as $\vec{X}$. So, $\vec{X}$ is not eigen vector of A.

Ans

b) Mathematical Analysis: for $\vec{x}\begin{pmatrix}2\\2\end{pmatrix}$

Given that,

$$A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}$$

let

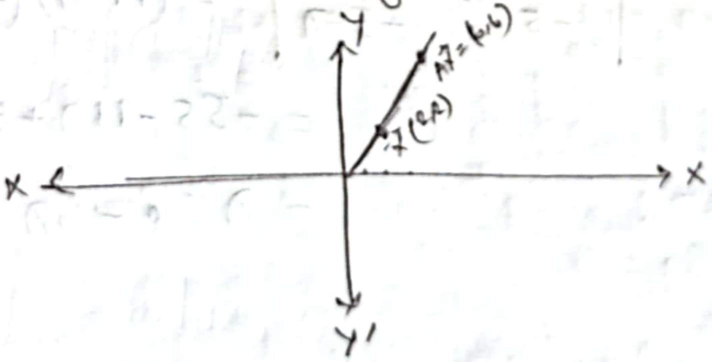
$$\vec{x} = \begin{pmatrix}2\\2\end{pmatrix}$$

$$A\vec{x} = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \cdot \begin{pmatrix}2\\2\end{pmatrix} = \begin{pmatrix}2+4\\-8+14\end{pmatrix} = \begin{pmatrix}6\\6\end{pmatrix} = 3\begin{pmatrix}2\\2\end{pmatrix}$$

As $A\vec{x} = \lambda\vec{x}$ is satisfied where $\lambda = 3$. So the given vector $\vec{x} = \begin{pmatrix}2\\2\end{pmatrix}$ is an Eigen vector. // (Ans)

b) Graphical Analysis:

Here, $A\vec{x} = \begin{pmatrix}6\\6\end{pmatrix}$ and $\vec{x} = \begin{pmatrix}2\\2\end{pmatrix}$



As the product vector, $A\vec{x}$ is in the same direction as $\vec{x}$. So, $\vec{x}$ is an Eigen vector of A. //

(Ans)

Q. Find the Eigen values and Eigen vectors of matrix $A = \begin{bmatrix} 11 & 3 \\ -5 & -5 \end{bmatrix}$

$\Rightarrow$ Given that,

$$A = \begin{bmatrix} 11 & 3 \\ -5 & -5 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 11 & 3 \\ -5 & -5 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 11 & 3 \\ -5 & -5 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} 11-\lambda & 3 \\ -5 & -5-\lambda \end{bmatrix}$$

So, the characteristic equation,

$$\therefore |A - \lambda I| = \begin{vmatrix} 11-\lambda & 3 \\ -5 & -5-\lambda \end{vmatrix} = (11-\lambda)(-5-\lambda) + 15$$

$$= -55 - 11\lambda + 5\lambda + \lambda^2 + 15$$

$$= \lambda^2 - 6\lambda - 40$$

$$|A - \lambda I| = 0$$

or, $\lambda^2 - 6\lambda - 40 = 0$

$\therefore \lambda_1 = 10$, and $\lambda_2 = -4$ (Ans)

Let a non-zero vector $\vec{v} = \begin{bmatrix} x \\ y \end{bmatrix}$

we have, $A\vec{v} = \lambda \vec{v}$

$$\begin{bmatrix}11 & 3\\ -5 & -5\end{bmatrix}\begin{bmatrix}x\\ y\end{bmatrix} = \lambda\begin{bmatrix}x\\ y\end{bmatrix}$$

or, $$\begin{bmatrix}11 & 3\\ -5 & -5\end{bmatrix}\begin{bmatrix}x\\ y\end{bmatrix} = -4\begin{bmatrix}x\\ y\end{bmatrix} \quad [\text{when } \lambda = -4]$$

or, $$\begin{bmatrix}11x+3y\\ -5x-5y\end{bmatrix} = -4\begin{bmatrix}x\\ y\end{bmatrix}$$

then, we obtain the system of equation,

$$11x+3y = -4x$$
$$-5x-5y = -4y$$

Now, $11x+3y=-4x$ or, $15x+3y=0$ $\therefore 5x+y=0$

And, $-5x-5y=-4y$ or, $-5x-y=0$

the system of equation,

$$5x+y=0 \quad \text{—}(1)$$
$$-5x-y=0 \quad \text{—}(2)$$

From eq^n (1) $\cancel{\Rightarrow x=0, \; y=0}$

$$\cancel{X = \begin{bmatrix}x\\ y\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix} = 0}$$

So, 1st and 2nd equations are identical. We can delete one of them,

$$-5x-y=0$$

In echelon form, there are only one equation in two unknow; then the system has a non-zero solution and in particular $(2-1)=1$ free variable, which is y,

let, $y=1$ $\therefore x=-1/5$

$\therefore V_1 = \begin{bmatrix} -1/5 \\ 1 \end{bmatrix}$, let $V_2 = \begin{bmatrix} x \\ y \end{bmatrix}$

We have $AV_2 = \lambda V_2$

or, $\begin{bmatrix} 11 & 3 \\ -5 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 10 \times \begin{bmatrix} x \\ y \end{bmatrix}$

or, $\begin{bmatrix} 11x + 3y \\ -5x - 5y \end{bmatrix} = \begin{bmatrix} 10x \\ 10y \end{bmatrix}$

Then we obtain the system of equations,

$11x + 3y = 10x$ and, $-5x - 5y = 10y$

$\therefore x + 3y = 0$ — ① $5x + 15y = 0$

$\therefore x + 3y = 0$ — ②

Since the 1st and 2nd equations are identical, we can delete one of them;

$x + 3y = 0$

In echelon form, there are only one equation in two unknown then the system has a non-zero solution and in particular $(2-1)=1$ free variable, which is y.

let $y = 1$ $\therefore x = -3$

$\therefore V_2 = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$

$\therefore P = [V_1, V_2] = \begin{bmatrix} -1/5 & -3 \\ 1 & 1 \end{bmatrix}$ which is eigen vector

(Ans)

Diagonalization of Matrix

Diagonal Matrix: A square matrix in which every element expect the principal diagonal elements is zero is called a Diagonal Matrix. Example: $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

Q. Find the eigen values and the corresponding eigen vectors of $A = \begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}$ and hence diagonalize the matrix.

⇨ **Formula:** $D = P^{-1} . A . P$

where, D is diagonal Matrix
A is given Matrix
P is the matrix of Eigen Vector
P^{-1} " " inverse matrix of P

Sol^n: Given that,

$$A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$$

$$\therefore A - \lambda I = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} 5-\lambda & 4 \\ 1 & 2-\lambda \end{bmatrix}$$

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So, the characteristic Equation,

$$|A - \lambda I| = \begin{vmatrix} 5-\lambda & 4 \\ 1 & 2-\lambda \end{vmatrix} = (5-\lambda)(2-\lambda) - 4$$

$$= 10 - 5\lambda - 2\lambda + \lambda^2 - 4$$

$$= \lambda^2 - 7\lambda + 6$$

Now,

$$|A - \lambda I| = 0$$

or, $\lambda^2 - 7\lambda + 6 = 0$ $\therefore \lambda = 1, 6$

Let a non-zero vector, $V = \begin{bmatrix} x \\ y \end{bmatrix}$

We have,

$$A\vec{V} = \lambda \vec{V}$$

or, $\begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 1 \begin{bmatrix} x \\ y \end{bmatrix}$ [when $\lambda = 1$]

or, $\begin{bmatrix} 5x + 4y \\ x + 2y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$

Then we obtain the system of Equation

$5x + 4y = x$	and $x + 2y = y$
or, $4x + 4y = 0$	$\therefore x + y = 0$ —②
$\therefore x + y = 0$ —①	

So, the 1st and 2nd Equations are indentical, We can delete one of them,

$$x + y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular $(2-1)=1$ free variable, which is y.

let,

$$y=1 \therefore x=-1$$

$$\therefore v_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad \text{let } v_2 = \begin{bmatrix} x \\ y \end{bmatrix}$$

We have $Av_2 = \lambda v_2$

$$\text{or, } \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = 6 \begin{bmatrix} x \\ y \end{bmatrix} \quad [\text{when } \lambda = 6]$$

$$\text{or, } \begin{bmatrix} 5x+4y \\ x+2y \end{bmatrix} = \begin{bmatrix} 6x \\ 6y \end{bmatrix}$$

Then we obtain the system of equations,

$$5x+4y = 6x \quad \text{and} \quad x+2y=6y$$

$$\text{or, } x-4y=0 \text{ — (1)} \quad \Big| \quad \text{or, } x-4y=0 \text{ — (2)}$$

Since the 1st and 2nd equations are identical. We can delete one of them;

$$x-4y=0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular $(2-1)=1$ free variable, which is y

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Let $y = 1$, $\therefore x = 4$

$$V_2 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

$$\therefore P = [V_1, V_2] = \begin{bmatrix} -1 & 4 \\ 1 & 1 \end{bmatrix}$$

$$|P| = \begin{vmatrix} -1 & 4 \\ 1 & 1 \end{vmatrix} = -1-4 = -5$$

Therefore,

$$P^{-1} = \frac{1}{|P|} \operatorname{adj}(P)$$

$$= \frac{1}{-5}\cancel{\begin{bmatrix} -1 & -1 \\ -4 & 1 \end{bmatrix}} = \frac{1}{-5}\begin{bmatrix} 1 & -4 \\ -1 & -1 \end{bmatrix}$$

$$= \cancel{\begin{bmatrix} 1/5 & 1/5 \\ -4/5 & -1/5 \end{bmatrix}} = \begin{bmatrix} -1/5 & 4/5 \\ 1/5 & 1/5 \end{bmatrix}$$

Hence, the required diagonal matrix,

$$D = P^{-1} A P$$

$$= \begin{bmatrix} -1/5 & 4/5 \\ 1/5 & 1/5 \end{bmatrix} \times \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} \times \begin{bmatrix} -1 & 4 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 + 4/5 & \frac{-4}{5} + \frac{8}{5} \\ 1 + 1/5 & \frac{4}{5} + \frac{2}{5} \end{bmatrix} \times \begin{bmatrix} -1 & 4 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{-1}{5} & 4/5 \\ \frac{6}{5} & 6/5 \end{bmatrix} \times \begin{bmatrix} -1 & 4 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1/5 + 4/5 & -4/5 + 4/5 \\ -6/5 + 6/5 & 24/5 + 6/5 \end{bmatrix}$$

$$= \begin{bmatrix} 9/5 & 0 \\ 0 & 6 \end{bmatrix}$$ // (Ans)

$\boxed{4}$. Find the area of the parallelogram defined by the column matrix,

$$u = \begin{bmatrix} -4 \\ 4 \end{bmatrix} \text{ and } v = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$

⇒ Given that,

$$u = \begin{bmatrix} -4 \\ 4 \end{bmatrix}, \; v = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$

~~det~~ $\therefore A = \begin{bmatrix} -4 & 6 \\ 4 & 2 \end{bmatrix}$

The Area of parallelogram is,

$$\therefore \det(A) = \begin{vmatrix} -4 & 6 \\ 4 & 2 \end{vmatrix} = |-8 - 24| = |-32|$$

$$= 32 \text{ square unit}$$

(Ans)

[illegible]. Prove that $A = \begin{bmatrix} 2 & 2-3i & 3+5i \\ 2+3i & 3 & i \\ 3-5i & -i & 5 \end{bmatrix}$ is Hermitian

⇨ Given that,

$$A = \begin{bmatrix} 2 & 2-3i & 3+5i \\ 2+3i & 3 & i \\ 3-5i & -i & 5 \end{bmatrix}$$

$$A = (\bar{A})^T$$

Conjugate Matrix of A ($\bar{A}$) :

$$\bar{A} = \begin{bmatrix} 2 & 2+3i & 3-5i \\ 2-3i & 3 & -i \\ 3+5i & i & 5 \end{bmatrix}$$

Transposed of Conjugate Matrix $\bar{A}$ $\{(\bar{A})^T\}$;

$$(\bar{A})^T = \begin{bmatrix} 2 & 2-3i & 3+5i \\ 2+3i & 3 & i \\ 3-5i & -i & 5 \end{bmatrix}$$

$\therefore A = (\bar{A})^T$ $\therefore A = \begin{bmatrix} 2 & 2-3i & 3+5i \\ 2+3i & 3 & i \\ 3-5i & -i & 5 \end{bmatrix}$ is Herm

(Proved)

4. Define ~~of~~ Matrix. If $A = [a_{ij}]$, where $a_{ij} = \begin{cases} 0, & \text{when } i \neq j \\ @, & \text{when } i = j \end{cases}$
then construct an 3x3 order matrix and Identify the type of Matrix. Also test the matrix A is orthogonal or not.
where @ is the sum of the 1st Digit and last Digit of your ID.

⇒ Here,

$$a_{ij} = \begin{cases} 0, & \text{when } i \neq j \\ 5, & \text{when } i = j \end{cases}$$

Matrix: Matrix is a set of numbers arranged in columns that forms a rectangular array.

$$\therefore A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$ //

It is a diagonal matrix. Because this matrix diagonal elements are non zero and other elements are equal to zero. //

If $A.A^T = I$, Then the matrix is considered orthogonal.

$$\therefore AA^T = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} \times \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 25 & 0 & 0 \\ 0 & 25 & 0 \\ 0 & 0 & 25 \end{bmatrix} \neq I$$

$\because A.A^T \neq I$. Hence the matrix A is not orthogonal.

// (Ans)